NAVTTECH-NEXSKILL – Batch,2026- Data Analytics

Instructions for Students: As a Data Analyst, read the business requirements carefully. Decide which statistical, logical, or aggregation function fits the scenario best, and apply it to extract the insights.

Q1. The Baseline Gross An investor wants to know the total box office business generated by all the top movies combined. Calculate the single aggregated figure for the entire gross revenue.

- **Dataset:** IMDb Top 1000

Q2. Record Count During a data audit, we need to confirm exactly how many transactions have been entered into our system. Looking at the 'OrderID' column, calculate the total number of entries.

- **Dataset:** Sales Dataset

Q3. The Extremes In the Superstore business, we need to identify our extreme financial outcomes from single transactions. Find the largest financial loss (highest negative profit) and the highest gain (maximum positive profit) in the dataset.

- **Dataset:** Superstore

Q4. The Typical Quantity The warehouse manager wants to know the most typical 'Quantity' ordered by customers at one time (the most common order size) so they can prepare packaging boxes accordingly.

- **Dataset:** Superstore

Q5. The Audit Sample The QA team needs to manually audit every 4th order out of the 1000 records. Apply a mathematical logic to the 'Row ID' that evaluates the number and returns a value allowing the QA team to easily filter only the 4th, 8th, 12th (and so on) rows.

- **Dataset:** Superstore

LEVEL 2: MEDIUM SCENARIOS (Conditional Logic & Ranking)

Q6. Category Performance The Sales Manager does not want the overall revenue. Calculate the total revenue strictly for items sold within the "Electronics" category.

- **Dataset:** Sales Dataset

Q7. The Bronze Medalist & The Shortest Everyone knows the top-grossing movie and the longest movie. Find the 'Gross' revenue of the movie that ranks exactly 3rd highest on the list, and find the runtime of the movie that ranks 5th from the bottom.

- **Dataset:** IMDb Top 1000

Q8. The True Middle Ground Our average sales are often heavily skewed by a few massive B2B orders. Ignoring the standard average, find the exact center point of the 'Sales' data where exactly half the orders fall below this value and half fall above it.

- **Dataset:** Superstore

Q9. Basic Tiering Logic Create a new column that evaluates the 'Profit'. If a transaction's profit is greater than 0, the cell should output "Profitable". If it is less than 0, it should output "Loss".

- **Dataset:** Superstore

Q10. Targeted Marketing We need to send a new campaign email to customers who live in the "South" region **AND** who have availed a 'Discount' greater than 0.2. Calculate the exact count of orders that meet both conditions simultaneously.

- **Dataset:** Superstore

LEVEL 3: COMPLEX SCENARIOS (Advanced Analysis & Nested Logic)

Q11. Multi-Condition Aggregation The VP of Sales has a highly specific request: Calculate the Total Profit and the Average Profit specifically for "Technology" items, but strictly if they were sold to the "Consumer" segment **AND** the order 'Quantity' was strictly greater than 5.

- **Dataset:** Superstore

Q12. The Volatility Check Investors want to understand the fluctuation in movie 'Gross' revenues. Measure the spread of the data to determine how unpredictable or far revenues tend to deviate from their overall average.

- **Dataset:** IMDb Top 1000

Q13. Strict B2B Promotions A special promotional gift will be dispatched to orders that are EITHER in the "Corporate" segment OR have an order amount greater than 1000. **HOWEVER**, if an order is both corporate and over 1000, it is disqualified from this specific gift. Write a logical formula to flag these exact orders.

- **Dataset:** Superstore

Q14. Rating Tiers Convert the numerical 'IMDB_Rating' column into text-based tags based on the following rules:

- If rating >= 9, output "Masterpiece"
- If >= 8, output "Excellent"
- If >= 7, output "Good"
- Everything else should be tagged as "Average".
- **Dataset:** IMDb Top 1000

Q15. The "Error-Proof" Dynamic Dashboard Create a Total cell for the 'Sales' column that updates dynamically when a user applies filters to the City or Region (meaning it automatically excludes hidden rows). Secure this entire system so that if the 'Sales' column is accidentally deleted in the future, the cell displays the clean text "System Error" instead of Excel's default `#REF!` error.

- **Dataset:** Superstore

LEVEL 4: EXPERT / BONUS SCENARIOS (Interactive BI Tasks)

Q16. The Elite Thresholds The Finance team needs to establish sales tiers. Analyze the data to find the exact threshold amount where the bottom 25% of our orders end, and the threshold where the top 25% of our orders begin. Finally, determine the cutoff value required to isolate strictly the top 5% VIP orders.

- **Dataset:** Superstore

Q17. The Region Coding The dataset contains full region names ("East", "West", "North", "South"). Using logic within a single cell, convert these names into single-letter codes (E, W, N, S). If any other text or spelling mistake appears in the row, it should automatically output the word "Review".

- **Dataset:** Superstore

Q18. Dynamic Positioning Evaluate every single 'Sales' value in the dataset and assign its exact numerical ranking position (e.g., 1 for the highest, 2 for the second highest, etc.) next to it, allowing the sales team to see where each order stands compared to the entire column.

- **Dataset:** Superstore

Q19. Exclusion Logic The Operations Manager needs to know how many total transactions were shipped via "Standard Class" mode. However, there is a strict condition: this count must **NOT** include any items belonging to the "Furniture" category. Calculate the final count.

- **Dataset:** Superstore